

Client Implementation Review

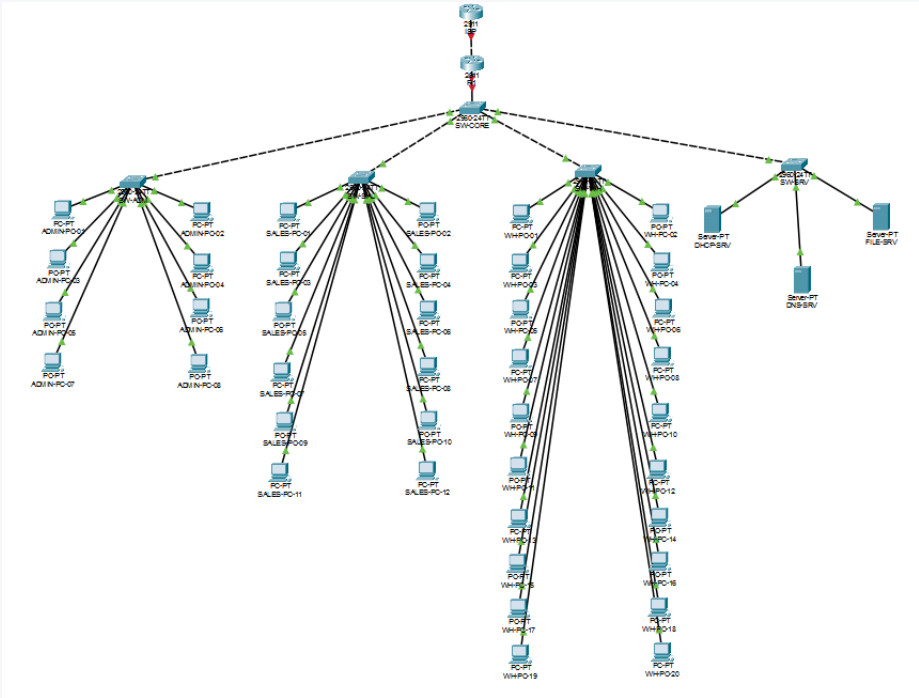
MILESTONE 2 · TSHIMO AGRI SUPPLIES NETWORK IMPLEMENTATION

Potchefstroom HQ, Agriculture

10.23.0.0/16

Fault Isolation — Advanced

12 / 12 tests passed



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PROJECT ID CMPG325-2026-042	CLIENT ID CLI-042
ASSIGNED CLIENT Tshimo Agri Supplies	ASSIGNED CHALLENGE Network Troubleshooting

- 1 Implementation Overview**
 What was built, on which devices, and how Milestone 2's requirements are met
- 2 Physical Build & Cabling**
 Device placement, port map, cable types and the corrections made during the build
- 3 Switching — VLANs & Trunks**
 VLAN database, 802.1Q trunks, native VLAN 99, access ports, management SVIs
- 4 Routing, WAN & NAT**
 Router-on-a-stick on R1, default route to the ISP, NAT/PAT overload
- 5 Network Services**
 Central DHCP with relay, internal DNS, and the file/web server
- 6 Testing & Evidence**
 Twelve functional tests run in Packet Tracer, with the captured results
- 7 Assigned Challenge — Advanced Fault Isolation**
 A VLAN pruned from a trunk: baseline, symptoms, layered isolation, root cause, fix, verification
- 8 Issues Encountered & Resolutions**
 Problems hit during implementation, how each was detected, and how it was fixed
- 9 Change Request & Growth Readiness**
 CR1 implemented on the live network; branch-office space still untouched
- 10 GitHub Portfolio**
 Repository structure and commit history for this milestone
- A Appendix — Device Configurations**
 The IOS command sets applied to every switch and router, verbatim

NETWORK DEVICES

7

END DEVICES

40 PCs + 3 servers

PHYSICAL LINKS

49

VLANS

10 · 20 · 30 · 40 · 99

FUNCTIONAL TESTS

12 / 12 passed

ASSIGNED FAULT

Isolated & repaired

IMPLEMENTATION APPROACH

The Milestone 1 design was built without changing a single VLAN, subnet or gateway. Every configuration was written as a reviewable command file first, applied one device at a time, and checked with a **show** command before moving on. Every claim in this document is backed by a screenshot taken from the running Packet Tracer network.

Milestone 2 turns the approved Milestone 1 design into a working network in Cisco Packet Tracer 9.0.1, then proves it works and demonstrates the assigned troubleshooting challenge on it.

1.1 Milestone 2 requirements and where each is met

REQUIREMENT	DELIVERED AS	WHERE
Working Packet Tracer file	Complete HQ network: 7 network devices, 43 end devices, all configured and saved in the repaired, healthy state	tshimo-agri-hq-milestone-2.pkt
Assigned feature implemented	Advanced fault isolation: a VLAN pruned from a trunk, injected, diagnosed layer by layer, fixed and verified	Section 7
Testing evidence	12 functional tests across Layers 1–7, every one captured from the live simulation	Sections 3–6
Updated GitHub portfolio	Configs, screenshots, test log, troubleshooting record and this report, committed in logical steps — github.com/MALULEKE-KS/tshimo-agri-network	Section 10

1.2 Device inventory

DEVICE	PACKET TRACER MODEL	ROLE
ISP	Cisco 2911	Simulated provider: far end of the WAN link, loopback 203.0.113.1 as the "internet" test target
R1	Cisco 2911	Edge router: inter-VLAN routing (router-on-a-stick), DHCP relay, default route, NAT/PAT
SW-CORE	Cisco 2960-24TT	Layer 2 distribution: trunks R1 down to all four access switches
SW-ADM / SW-SAL / SW-WH / SW-SRV	Cisco 2960-24TT	One access switch per department; 24 ports each for physical headroom
ADMIN-PC-01...08	PC-PT	VLAN 10 Admin/Management users
SALES-PC-01...12	PC-PT	VLAN 20 Sales users
WH-PC-01...20	PC-PT	VLAN 30 Warehouse/Logistics users, including the 8 CR1 staff
DHCP-SRV / DNS-SRV / FILE-SRV	Server-PT	VLAN 40 Central DHCP, internal DNS, file and web services

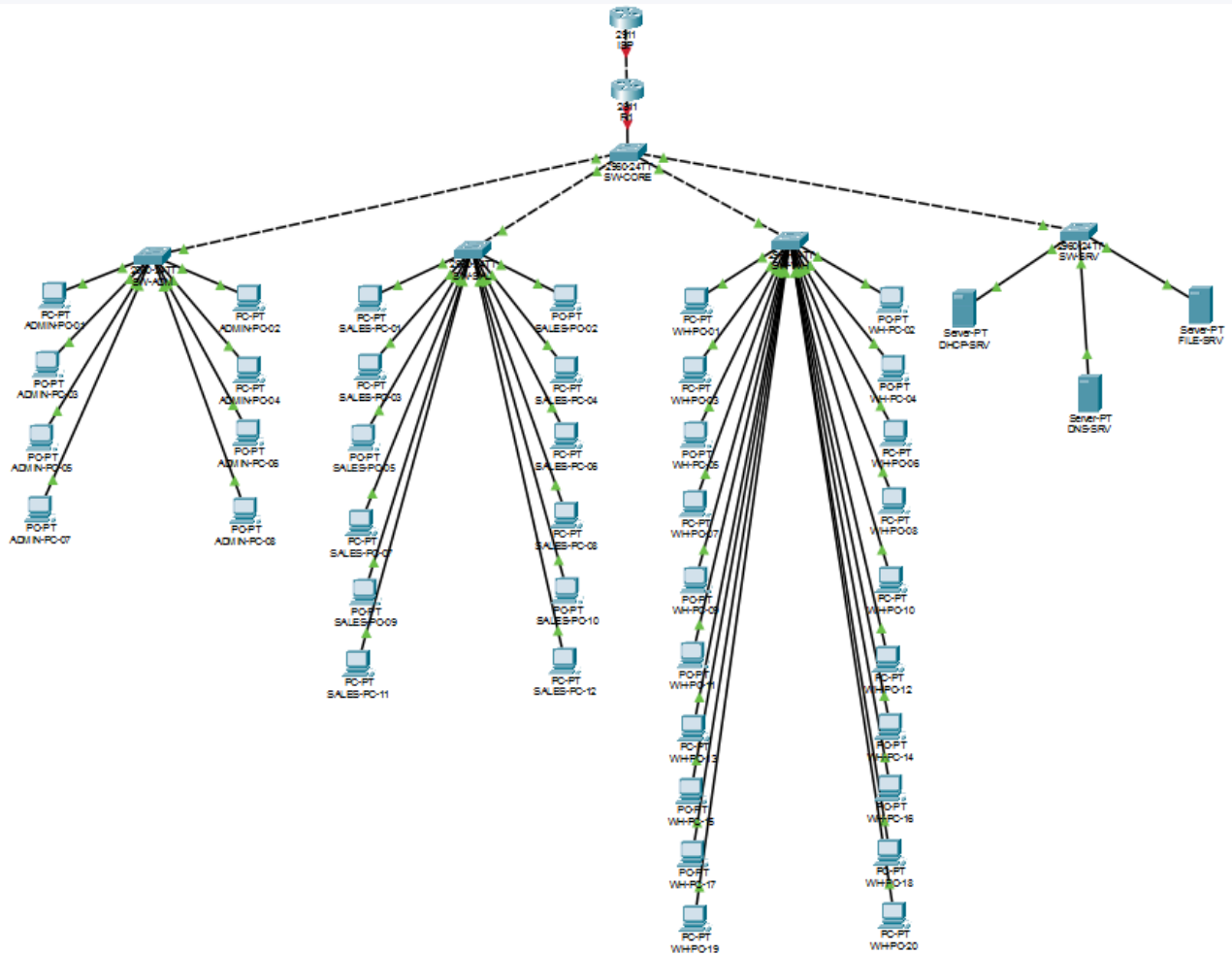


Figure 1. The implemented HQ topology in Packet Tracer: ISP → R1 → SW-CORE → four access switches → 40 PCs and 3 servers.

The physical build was completed and saved as an unconfigured baseline (`tshimo-agri-hq-baseline.pkt`) before any IOS configuration, so the wiring could be checked on its own.

2.1 Uplink and trunk cabling

CABLE	FROM	TO	CABLE TYPE	LOGICAL ROLE
C01	ISP Gi0/0	R1 Gi0/1	Crossover	Routed WAN, 10.23.1.0/30
C02	R1 Gi0/0	SW-CORE Gi0/1	Straight-through	802.1Q trunk (router-on-a-stick)
C03	SW-CORE Fa0/1	SW-ADM Fa0/1	Crossover	802.1Q trunk
C04	SW-CORE Fa0/2	SW-SAL Fa0/1	Crossover	802.1Q trunk
C05	SW-CORE Fa0/3	SW-WH Fa0/1	Crossover	802.1Q trunk
C06	SW-CORE Fa0/4	SW-SRV Fa0/1	Crossover	802.1Q trunk

Cable types follow the standard rule: **straight-through between unlike devices** (router↔switch, switch↔PC/server) and **crossover between like devices** (router↔router, switch↔switch). The 2911 and 2960 support Auto-MDIX and would usually bring a wrong cable up anyway, but the build uses the textbook-correct cable so it doesn't rely on that.

2.2 Endpoint ports (all straight-through)

SWITCH	PORTS	DEVICES	VLAN
SW-ADM	Fa0/2 - Fa0/9	8 Admin PCs	10
SW-SAL	Fa0/2 - Fa0/13	12 Sales PCs	20
SW-WH	Fa0/2 - Fa0/21	20 Warehouse PCs (12 + 8 CR1)	30
SW-SRV	Fa0/2, Fa0/3, Fa0/4	DHCP-SRV, DNS-SRV, FILE-SRV	40

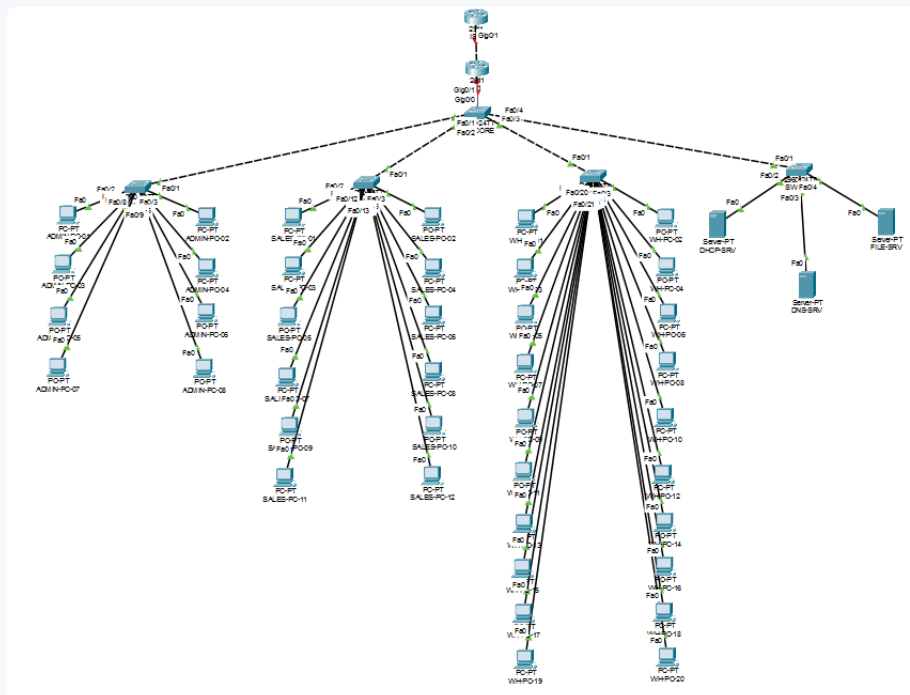


Figure 2. The same topology with Packet Tracer's port labels switched on. The four SW-CORE ↔ access-switch links are drawn dashed (crossover); endpoint links are solid (straight-through). The ISP-R1 link is too short at this zoom to show its dash pattern.

Two corrections made during the build — both are detailed in Section 8. (1) The original port plan put SW-CORE's four access uplinks on Gi0/2–Gi0/5, but a 2960-24TT only has two Gigabit ports, so those uplinks moved to Fa0/1–Fa0/4. (2) The five like-device links were first cabled straight-through and were recabled as crossover after a check against the rubric's "correct cabling" criterion. Neither change touched any VLAN, subnet or gateway.

3.1 VLAN and addressing plan as implemented

VLAN	NAME	SUBNET	GATEWAY (R1)	WHERE IT LIVES
10	ADMIN	10.23.0.0/26	10.23.0.1	SW-ADM access ports
20	SALES	10.23.0.64/26	10.23.0.65	SW-SAL access ports
30	WAREHOUSE	10.23.0.128/26	10.23.0.129	SW-WH access ports
40	SERVERS	10.23.0.192/26	10.23.0.193	SW-SRV access ports
99	MGMT-NATIVE	10.23.1.16/28	10.23.1.17	Native VLAN on every trunk; switch management SVIs

3.2 What each switch was configured with

- VLAN database.** SW-CORE holds all five VLANs, since every VLAN crosses it. Each access switch only creates the VLAN its ports use, plus VLAN 99.
- Trunks.** Every switch-to-switch and switch-to-router link is a static 802.1Q trunk (`switchport mode trunk`) with **native VLAN 99** and an explicit allowed list of `10,20,30,40,99` . Moving the native VLAN off VLAN 1 keeps untagged traffic out of the default VLAN.
- Access ports.** Each department's port range is a static access port in its VLAN, with PortFast so PCs come up without waiting for spanning tree.
- Management.** Each switch has an SVI on VLAN 99 — SW-CORE `.18` , SW-ADM `.19` , SW-SAL `.20` , SW-WH `.21` , SW-SRV `.22` (all /28) — with default gateway `10.23.1.17` .

```
SW-CORE#
SW-CORE#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/2 (99), with Switch FastEthernet0/1 (1).
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/3 (99), with Switch FastEthernet0/1 (1).
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (99), with Switch FastEthernet0/1 (1).
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/4 (99), with Switch FastEthernet0/1 (1).
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/2 (99), with Switch FastEthernet0/1 (1).
%SPANTREE-2-UNBLOCK_CONSIST_PORT: Unblocking FastEthernet0/1 on VLAN0001. Port consistency restored.
%SPANTREE-2-UNBLOCK_CONSIST_PORT: Unblocking FastEthernet0/1 on VLAN0099. Port consistency restored.

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/3 (99), with Switch FastEthernet0/1 (1).
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/4 (99), with Switch FastEthernet0/1 (1).
%SPANTREE-2-UNBLOCK_CONSIST_PORT: Unblocking FastEthernet0/4 on VLAN0001. Port consistency restored.
%SPANTREE-2-UNBLOCK_CONSIST_PORT: Unblocking FastEthernet0/4 on VLAN0099. Port consistency restored.

SW-CORE#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID        Local Intrfce    Holdtme    Capability    Platform    Port ID
SW-ADM           Fas 0/1          131        S             2960        Fas 0/1
SW-SAL           Fas 0/2          157        S             2960        Fas 0/1
SW-WH            Fas 0/3          178        S             2960        Fas 0/1
SW-SRV           Fas 0/4          135        S             2960        Fas 0/1
SW-CORE#
```

Figure 3. SW-CORE `show cdp neighbors` : Fa0/1 → SW-ADM, Fa0/2 → SW-SAL, Fa0/3 → SW-WH, Fa0/4 → SW-SRV, each on the neighbour's Fa0/1 — the port map confirmed from the switch itself. The native-VLAN-mismatch messages above it were logged while the switches were configured one at a time; each cleared ("Port consistency restored") once both ends of a trunk had native VLAN 99. R1 isn't listed yet because it was configured afterwards.

```

SW-CORE#show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/1     on        802.1q         trunking    99
Fa0/2     on        802.1q         trunking    99
Fa0/3     on        802.1q         trunking    99
Fa0/4     on        802.1q         trunking    99

Port      Vlans allowed on trunk
Fa0/1     10,20,30,40,99
Fa0/2     10,20,30,40,99
Fa0/3     10,20,30,40,99
Fa0/4     10,20,30,40,99

Port      Vlans allowed and active in management domain
Fa0/1     10,20,30,40,99
Fa0/2     10,20,30,40,99
Fa0/3     10,20,30,40,99
Fa0/4     10,20,30,40,99

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     10,20,30,40,99
Fa0/2     10,20,30,40
Fa0/3     10,20,30,40
--More--

```

Figure 4. SW-CORE `show interfaces trunk`: Fa0/1–Fa0/4 all trunking, 802.1Q, native VLAN 99, allowed 10,20,30,40,99. (Gi0/1 to R1 came up once R1 was configured — see Figure 17.)

```

SW-WH#show vlan brief
VLAN Name                Status    Ports
-----
1    default              active    Fa0/22, Fa0/23, Fa0/24, Gig0/1
                               Gig0/2
30   WAREHOUSE             active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                               Fa0/6, Fa0/7, Fa0/8, Fa0/9
                               Fa0/10, Fa0/11, Fa0/12, Fa0/13
                               Fa0/14, Fa0/15, Fa0/16, Fa0/17
                               Fa0/18, Fa0/19, Fa0/20, Fa0/21
99   MGMT-NATIVE           active
1002 fddi-default         active
1003 token-ring-default   active
1004 fddinet-default       active
1005 trnet-default        active
SW-WH#

```

Figure 5. SW-WH `show vLan brief`: VLAN 30 WAREHOUSE holds exactly 20 access ports, Fa0/2–Fa0/21 — the 12 original Warehouse staff plus the 8 added by Change Request 1.

4.1 Router-on-a-stick on R1

The switches are Layer 2 only, so R1 is the single point where traffic moves between VLANs. Its Gi0/0 carries no IP address; instead it has one 802.1Q subinterface per VLAN, and each subinterface's address is that VLAN's default gateway.

SUBINTERFACE	ENCAPSULATION	IP ADDRESS	DHCP RELAY	NAT
Gi0/0.10	dot1Q 10	10.23.0.1/26	10.23.0.194	inside
Gi0/0.20	dot1Q 20	10.23.0.65/26	10.23.0.194	inside
Gi0/0.30	dot1Q 30	10.23.0.129/26	10.23.0.194	inside
Gi0/0.40	dot1Q 40	10.23.0.193/26	— (servers are static)	inside
Gi0/0.99	dot1Q 99 native	10.23.1.17/28	—	—
Gi0/1	— (WAN)	10.23.1.1/30	—	outside

`encapsulation dot1Q 99 native` matches the switches' native VLAN, so untagged management frames on the trunk land in the right subinterface.

4.2 Default route and NAT/PAT

- Default route:** `ip route 0.0.0.0 0.0.0.0 10.23.1.2` sends everything not local to the ISP.
- NAT/PAT overload:** `access-list 1 permit 10.23.0.0 0.0.0.255` + `ip nat inside source list 1 interface Gi0/1 overload`. Every user and server address shares R1's WAN address `10.23.1.1`, told apart by port number.
- Deliberately excluded:** the ACL only matches `10.23.0.0/24`. The WAN and management ranges (`10.23.1.0/24`) are never translated, so infrastructure traffic keeps its real addresses.
- ISP side:** Gi0/0 `10.23.1.2/30` and Loopback0 `203.0.113.1/32` (a documentation-range address used as the "internet" test target). The ISP needs no route back to the LAN: it only ever sees R1's translated address, which is directly connected.

```
R1#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0       unassigned      YES unset    up          up
GigabitEthernet0/0.10    10.23.0.1       YES manual    up          up
GigabitEthernet0/0.20    10.23.0.65      YES manual    up          up
GigabitEthernet0/0.30    10.23.0.129     YES manual    up          up
GigabitEthernet0/0.40    10.23.0.193     YES manual    up          up
GigabitEthernet0/0.99    10.23.1.17      YES manual    up          up
GigabitEthernet0/1       10.23.1.1       YES manual    up          up
GigabitEthernet0/2       unassigned      YES unset    administratively down down
Vlan1                    unassigned      YES unset    administratively down down
R1#
```

Figure 6. R1 `show ip interface brief`: all five VLAN subinterfaces and the WAN link are up/up with their planned addresses. (Gi0/2 is unused and stays shut.)

```
R1#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0       unassigned      YES unset    up          up
GigabitEthernet0/0.10    10.23.0.1       YES manual   up          up
GigabitEthernet0/0.20    10.23.0.65      YES manual   up          up
GigabitEthernet0/0.30    10.23.0.129     YES manual   up          up
GigabitEthernet0/0.40    10.23.0.193     YES manual   up          up
GigabitEthernet0/0.99    10.23.1.17      YES manual   up          up
GigabitEthernet0/1       10.23.1.1       YES manual   up          up
GigabitEthernet0/2       unassigned      YES unset    administratively down down
Vlan1                    unassigned      YES unset    administratively down down
R1#how ip route
^
% Invalid input detected at '^' marker.

R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.23.1.2 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 12 subnets, 4 masks
C       10.23.0.0/26 is directly connected, GigabitEthernet0/0.10
L       10.23.0.1/32 is directly connected, GigabitEthernet0/0.10
C       10.23.0.64/26 is directly connected, GigabitEthernet0/0.20
L       10.23.0.65/32 is directly connected, GigabitEthernet0/0.20
C       10.23.0.128/26 is directly connected, GigabitEthernet0/0.30
L       10.23.0.129/32 is directly connected, GigabitEthernet0/0.30
C       10.23.0.192/26 is directly connected, GigabitEthernet0/0.40
L       10.23.0.193/32 is directly connected, GigabitEthernet0/0.40
C       10.23.1.0/30 is directly connected, GigabitEthernet0/1
L       10.23.1.1/32 is directly connected, GigabitEthernet0/1
C       10.23.1.16/28 is directly connected, GigabitEthernet0/0.99
--More-- |
```

Figure 7. R1 `show ip route`: a connected (C) and local (L) route for every VLAN and for the WAN /30, and "Gateway of last resort is 10.23.1.2" confirming the default route. The "% Invalid input" line is a typo (`how ip route`) corrected on the next line.

5.1 Server addressing (static, VLAN 40)

SERVER	IP / MASK	GATEWAY	DNS	SERVICE ENABLED
DHCP-SRV	10.23.0.194/26	10.23.0.193	10.23.0.195	DHCP (pools below)
DNS-SRV	10.23.0.195/26	10.23.0.193	10.23.0.195	DNS — A record <code>files.tshimo.test → 10.23.0.196</code>
FILE-SRV	10.23.0.196/26	10.23.0.193	10.23.0.195	HTTP and FTP

5.2 Central DHCP with relay

POOL	DEFAULT GATEWAY	DNS	START ADDRESS	MASK	MAX USERS
ADMIN-POOL	10.23.0.1	10.23.0.195	10.23.0.10	255.255.255.192	50
SALES-POOL	10.23.0.65	10.23.0.195	10.23.0.74	255.255.255.192	50
WH-POOL	10.23.0.129	10.23.0.195	10.23.0.138	255.255.255.192	50
serverPool1	10.23.0.193	10.23.0.195	10.23.0.200	255.255.255.192	20

A PC's DHCP request is a broadcast and never leaves its own VLAN. R1's `ip helper-address 10.23.0.194` on each user subinterface picks it up and forwards it to DHCP-SRV as unicast, stamped with the subinterface's gateway address. The server uses that gateway to choose the matching pool. Each pool starts nine addresses above its gateway, which keeps `.1-.9` of every /26 free for infrastructure. All 40 user PCs were set to DHCP. Leases were checked on one PC per department (Figure 8), and three more PCs were used successfully in later tests.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig /all

FastEthernet0 Connection: (default port)

Connection-specific DNS Suffix...:
Physical Address. . . . .: 0009.70C6.25A6
Link-local IPv6 Address . . . . .: FE80::209:7CFF:FE6:25A6
IPv6 Address. . . . .: ::
IPv4 Address. . . . .: 10.23.0.74
Subnet Mask . . . . .: 255.255.255.192
Default Gateway. . . . .: ::
                        10.23.0.65
DHCP Servers. . . . .: 10.23.0.194
DHCPv6 IAID. . . . .:
DHCPv6 Client DUID. . . . .: 00-01-00-01-61-5D-8A-20-00-09-7C-C6-25-A6
DNS Servers. . . . .: 10.23.0.195
```

Figure 8b. SALES-PC-01: `10.23.0.74/26`, gateway `.65` — first SALES-POOL lease.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig /all

FastEthernet0 Connection: (default port)

Connection-specific DNS Suffix...:
Physical Address. . . . .: 000A.F392.AD43
Link-local IPv6 Address . . . . .: FE80::20A:F3FF:FE92:AD43
IPv6 Address. . . . .: ::
IPv4 Address. . . . .: 10.23.0.138
Subnet Mask . . . . .: 255.255.255.192
Default Gateway. . . . .: ::
                        10.23.0.129
DHCP Servers. . . . .: 10.23.0.194
DHCPv6 IAID. . . . .:
DHCPv6 Client DUID. . . . .: 00-01-00-01-61-5D-8A-20-00-0A-F3-92-AD-43
DNS Servers. . . . .: 10.23.0.195
```

Figure 8c. WH-PC-01: `10.23.0.138/26`, gateway `.129` — first WH-POOL lease.

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig /all

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Physical Address.....: 0007.EC20.E34B
    Link-local IPv6 Address.....: FE80::207:ECFF:FE20:E34B
    IPv6 Address.....: ::
    IPv4 Address.....: 10.23.0.10
    Subnet Mask.....: 255.255.255.192
    Default Gateway.....: ::
                        10.23.0.1

    DHCP Servers.....: 10.23.0.194
    DHCPv6 IAID.....:
    DHCPv6 Client DUID.....: 00-01-00-01-61-5D-8A-20-00-07-EC-20-E3-4B
    DNS Servers.....: ::
                        10.23.0.195

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Physical Address.....: 0001.644D.8BD5
    Link-local IPv6 Address.....: ::
    IPv6 Address.....: ::
    IPv4 Address.....: 0.0.0.0
    Subnet Mask.....: 0.0.0.0
    Default Gateway.....: ::
                        0.0.0.0

    DHCP Servers.....: 0.0.0.0
    DHCPv6 IAID.....:
    DHCPv6 Client DUID.....: 00-01-00-01-61-5D-8A-20-00-07-EC-20-E3-4B

C:\>ping 10.23.0.196

Pinging 10.23.0.196 with 32 bytes of data:

Request timed out.
Reply from 10.23.0.196: bytes=32 time<1ms TTL=127
Reply from 10.23.0.196: bytes=32 time<1ms TTL=127
Reply from 10.23.0.196: bytes=32 time<1ms TTL=127

Ping statistics for 10.23.0.196:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

```

Figure 8a. ADMIN-PC-01 `ipconfig /all`: `10.23.0.10/26`, gateway `10.23.0.1`, DHCP server `10.23.0.194`, DNS `10.23.0.195`.
The ping that follows (VLAN 10 → file server in VLAN 40) gets 3 of 4 replies; the first is lost while ARP resolves, which is normal.

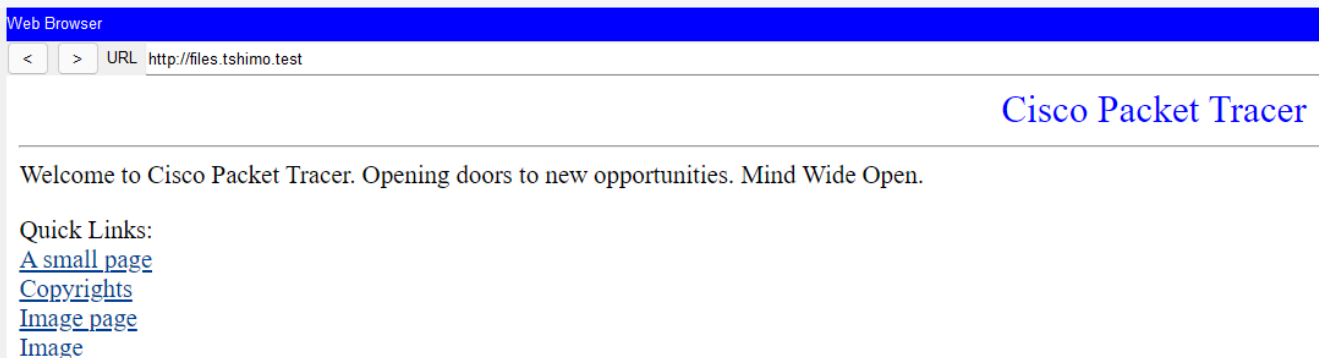


Figure 9. ADMIN-PC-04's web browser opening `http://files.tshimo.test`; the name resolves via DNS-SRV, and FILE-SRV serves the page.

Every test was run live in Packet Tracer against the configured network. Observed results come only from what the simulation showed.

ID	TEST	OBSERVED	RESULT	FIG.
PHY-01	Topology and cabling	All 7 network devices, 43 end devices and 49 links present; crossover on like-device links	PASS	1, 2
L2-00	SW-CORE <code>show cdp neighbors</code>	Fa0/1–Fa0/4 → SW-ADM, SW-SAL, SW-WH, SW-SRV as planned	PASS	3
L2-01	SW-WH <code>show vlan brief</code>	VLAN 30 active on Fa0/2–Fa0/21	PASS	5
L2-02	SW-CORE <code>show interfaces trunk</code>	All trunks: 802.1Q, native 99, allowed 10,20,30,40,99	PASS	4
L3-01	R1 <code>show ip interface brief</code>	5 subinterfaces + Gi0/1 up/up with planned addresses	PASS	6
L3-02	R1 <code>show ip route</code>	All VLAN and WAN routes connected; default via 10.23.1.2	PASS	7
DHCP-01	One PC per user VLAN, <code>ipconfig /all</code>	Each got the first address of its own pool, correct gateway and DNS	PASS	8a–c
LAN-01	WH-PC-01 → WH-PC-02 (same VLAN)	4/4 replies	PASS	10
ROUTE-01	Admin and Sales → FILE-SRV (across VLANs)	ADMIN-PC-01 3/4 (first lost to ARP), SALES-PC-01 4/4	PASS	8a, 11
DNS-01	Browser → <code>files.tshimo.test</code>	Name resolved, page loaded	PASS	9
NAT-01	PC → <code>203.0.113.1</code> through NAT	SALES-PC-03 4/4 replies, 0% loss	PASS	12, 13
CR1-01	Warehouse capacity after CR1	20 ports in VLAN 30, no subnet, gateway or trunk change	PASS	5

ROUTE-01 note: Warehouse's own path to the file server is demonstrated in Section 7 (Figure 14, healthy baseline: 4/4).

```
C:\>ping 10.23.0.139
Pinging 10.23.0.139 with 32 bytes of data:
Reply from 10.23.0.139: bytes=32 time=27ms TTL=128
Reply from 10.23.0.139: bytes=32 time<1ms TTL=128
Reply from 10.23.0.139: bytes=32 time<1ms TTL=128
Reply from 10.23.0.139: bytes=32 time<1ms TTL=128
Ping statistics for 10.23.0.139:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 27ms, Average = 6ms
```

Figure 10. WH-PC-01 → WH-PC-02 (`10.23.0.139`), same VLAN: 4/4.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 203.0.113.1
Pinging 203.0.113.1 with 32 bytes of data:
Reply from 203.0.113.1: bytes=32 time=47ms TTL=254
Reply from 203.0.113.1: bytes=32 time<1ms TTL=254
Reply from 203.0.113.1: bytes=32 time<1ms TTL=254
Reply from 203.0.113.1: bytes=32 time<1ms TTL=254
Ping statistics for 203.0.113.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 47ms, Average = 11ms
```

Figure 12. SALES-PC-03 → `203.0.113.1` (ISP loopback) through NAT: 4/4, 0% loss.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig /all

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Physical Address.....: 0009.7CC6.25A6
    Link-local IPv6 Address.....: FE80::209:7CFF:FEC6:25A6
    IPv6 Address.....: ::
    IPv4 Address.....: 10.23.0.74
    Subnet Mask.....: 255.255.255.192
    Default Gateway.....: ::
    DHCP Servers.....: 10.23.0.65
    DHCPv6 IAID.....: 10.23.0.194
    DHCPv6 Client DUID.....: 00-01-00-01-61-5D-8A-20-00-09-7C-C6-25-A6
    DNS Servers.....: ::
    DNS Servers.....: 10.23.0.195

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Physical Address.....: 0090.212D.27E5
    Link-local IPv6 Address.....: ::

C:\>ping 10.23.0.196

Pinging 10.23.0.196 with 32 bytes of data:

Reply from 10.23.0.196: bytes=32 time=75ms TTL=127
Reply from 10.23.0.196: bytes=32 time<1ms TTL=127
Reply from 10.23.0.196: bytes=32 time<1ms TTL=127
Reply from 10.23.0.196: bytes=32 time<1ms TTL=127

Ping statistics for 10.23.0.196:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 75ms, Average = 18ms
```

Figure 11. SALES-PC-01 (VLAN 20, 10.23.0.74) → FILE-SRV (VLAN 40, 10.23.0.196): 4/4. TTL 127 instead of 128 shows the packet crossed one router hop (R1).

```
R1(config)#! --- Default route out to the ISP ---
R1(config)#ip route 0.0.0.0 0.0.0.0 10.23.1.2
R1(config)#
R1(config)#! --- NAT/PAT: translate the business/server range (10.23.0.0/24) only.
R1(config)#!      WAN and management (10.23.1.0/24) are deliberately excluded. ---
R1(config)#access-list 1 permit 10.23.0.0 0.0.0.255
R1(config)#ip nat inside source list 1 interface gigabitEthernet0/1 overload
R1(config)#
R1(config)#end
R1#write memory
Building configuration...
[OK]
R1#
R1#! Verify with:
R1#!   show ip interface brief           (all subinterfaces + Gi0/1 up/up)
R1#!   show ip route                   (C routes for each VLAN, S* default to 10.23.1.2)
R1#!   show ip nat translations        (after a PC pings 203.0.113.1)
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#show ip nat translations
R1#show ip nat translations
R1#
R1#
R1#
```

Figure 13. R1's NAT configuration as applied, followed by `show ip nat translations`. The table is empty because it was checked about six minutes after the NAT ping, and dynamic PAT entries expire after a short idle time. The successful ping in Figure 12 is the evidence that translation worked: an untranslated 10.23.0.x source could not have got a reply back from the ISP.

7.1 The scenario

A change on the core switch removes VLAN 30 (Warehouse) from the allowed-VLAN list on the SW-CORE → SW-WH trunk. It's a realistic one-word mistake: typing `switchport trunk allowed vlan 10,20,40,99` instead of `...10,20,30,40,99`, or running `remove 30` on the wrong port.

Why it qualifies as Advanced:

- 1 The physical link stays up. Link lights and Layer 1 checks look healthy, so the fault can't be seen on the topology.
- 2 The failure is partial. Warehouse PCs can still reach each other, which can mislead a troubleshooter into blaming the gateway, DHCP or the PCs.
- 3 The two ends of the trunk disagree: SW-WH still allows VLAN 30 and SW-CORE doesn't. Checking only one end misses it.
- 4 It's a one-line mistake with a one-line fix. Finding it depends on method, not on rebuilding anything.

7.2 Method

Work up the OSI model and only move up a layer once the one below is proven good. Use a **control test** from an unaffected department to scope the fault before touching any device. Treat expected symptoms as hypotheses to check, not conclusions.

7.3 Step 1 — Healthy baseline

```
C:\>ping 10.23.0.139

Pinging 10.23.0.139 with 32 bytes of data:

Reply from 10.23.0.139: bytes=32 time=27ms TTL=128
Reply from 10.23.0.139: bytes=32 time<1ms TTL=128
Reply from 10.23.0.139: bytes=32 time<1ms TTL=128
Reply from 10.23.0.139: bytes=32 time<1ms TTL=128

Ping statistics for 10.23.0.139:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 27ms, Average = 6ms

C:\>clear
Invalid Command.

C:\>ping 10.23.0.129

Pinging 10.23.0.129 with 32 bytes of data:

Reply from 10.23.0.129: bytes=32 time=46ms TTL=255
Reply from 10.23.0.129: bytes=32 time<1ms TTL=255
Reply from 10.23.0.129: bytes=32 time<1ms TTL=255
Reply from 10.23.0.129: bytes=32 time=209ms TTL=255

Ping statistics for 10.23.0.129:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 209ms, Average = 63ms

C:\>ping 10.23.0.196

Pinging 10.23.0.196 with 32 bytes of data:

Reply from 10.23.0.196: bytes=32 time=1ms TTL=127
Reply from 10.23.0.196: bytes=32 time<1ms TTL=127
Reply from 10.23.0.196: bytes=32 time<1ms TTL=127
Reply from 10.23.0.196: bytes=32 time<1ms TTL=127

Ping statistics for 10.23.0.196:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Figure 14. Before the fault, WH-PC-01 (10.23.0.138) reaches WH-PC-02 (.139), its gateway (.129) and FILE-SRV (.196): all 4/4. The Packet Tracer file was saved at this point.

7.4 Step 2 — Inject the fault (on SW-CORE, not saved)

```
configure terminal
interface fastEthernet0/3
  switchport trunk allowed vlan remove 30
end
! no write memory — the fault is never saved
```

7.5 Step 3 — Observe the symptoms

TEST	EXPECTED (HYPOTHESIS)	OBSERVED
WH-PC-01 → gateway (.129)	Fail: path to R1 crosses the trunk	100% loss
WH-PC-01 → FILE-SRV (.196)	Fail: another VLAN, needs R1	100% loss
WH-PC-01 → WH-PC-02 (.139)	Work: same switch, same VLAN, no trunk needed	4/4
ADMIN-PC-01 → FILE-SRV (control)	Work: Admin uses Fa0/1, not Fa0/3	4/4

```
C:\>ping 10.23.0.129
Pinging 10.23.0.129 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.23.0.129:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.23.0.139
Pinging 10.23.0.139 with 32 bytes of data:
Reply from 10.23.0.139: bytes=32 time=145ms TTL=128
Reply from 10.23.0.139: bytes=32 time<1ms TTL=128
Reply from 10.23.0.139: bytes=32 time=1ms TTL=128
Reply from 10.23.0.139: bytes=32 time<1ms TTL=128

Ping statistics for 10.23.0.139:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 145ms, Average = 36ms

C:\>ping 10.23.0.196
Pinging 10.23.0.196 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.23.0.196:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

```
C:\>ping 10.23.0.196
Pinging 10.23.0.196 with 32 bytes of data:
Reply from 10.23.0.196: bytes=32 time<1ms TTL=127
Reply from 10.23.0.196: bytes=32 time<1ms TTL=127
Reply from 10.23.0.196: bytes=32 time<1ms TTL=127
Reply from 10.23.0.196: bytes=32 time<1ms TTL=127

Ping statistics for 10.23.0.196:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure 16. ADMIN-PC-01 control test during the same fault: FILE-SRV answers 4/4, 0% loss.

Figure 15. WH-PC-01 during the fault: gateway (.129) and file server (.196) fail; WH-PC-02 (.139) still answers.

What the two results prove together. The control test passes and Warehouse fails, so the fault is scoped to **Warehouse only**. Everything Admin depends on (R1, SW-CORE's other ports, SW-SRV, the file server) is working. What's left to suspect is the path that only Warehouse uses: SW-WH and its uplink to SW-CORE.

7.6 Step 4 — Isolate layer by layer

LAYER	EVIDENCE	CONCLUSION
Scope	Admin works; Warehouse PCs reach each other	R1, the servers and SW-WH's access ports are fine. The problem lies on the path between SW-WH and R1.
1 — Physical	Link lights up; WH-PC-01 ↔ WH-PC-02 works	Cables and ports are fine.
3 — Network	WH-PC-01 still holds its DHCP lease 10.23.0.138/26, gateway .129; R1's Gi0/0.30 was up/up before the change	Not an addressing or gateway fault.
2 — Data link	SW-CORE show interfaces trunk (Figure 17)	All five trunk ports up, native 99. Fa0/3's allowed list reads 10,20,40,99; every other port reads 10,20,30,40,99. Fault found.

Root cause. SW-CORE Fa0/3 is the only physical path between SW-WH and the rest of the network. With VLAN 30 removed from that trunk's allowed list, no VLAN 30 frame can cross it. Warehouse PCs can still switch locally, but they lose everything that depends on reaching R1. Mode, encapsulation and native VLAN are all correct, so the fault is purely the trunk's VLAN membership.

```
SW-CORE#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
SW-CORE(config)#interface fastEthernet0/3
SW-CORE(config-if)# switchport trunk allowed vlan remove 30
SW-CORE(config-if)#end
SW-CORE#
%SYS-S-CONFIG_I: Configured from console by console

SW-CORE#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
SW-CORE(config)#interface fastEthernet0/3
SW-CORE(config-if)# switchport trunk allowed vlan remove 30
SW-CORE(config-if)#end
SW-CORE#
%SYS-S-CONFIG_I: Configured from console by console

SW-CORE#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
SW-CORE(config)#interface fastEthernet0/3
SW-CORE(config-if)# switchport trunk allowed vlan remove 30
SW-CORE(config-if)#end
SW-CORE#
%SYS-S-CONFIG_I: Configured from console by console

SW-CORE#show interfaces trunk
Port      Mode      Encapsulation  Status        Native vlan
Fa0/1     on        802.1q         trunking      99
Fa0/2     on        802.1q         trunking      99
Fa0/3     on        802.1q         trunking      99
Fa0/4     on        802.1q         trunking      99
Gig0/1    on        802.1q         trunking      99

Port      Vlans allowed on trunk
Fa0/1     10,20,30,40,99
Fa0/2     10,20,30,40,99
Fa0/3     10,20,40,99
Fa0/4     10,20,30,40,99
Gig0/1    10,20,30,40,99

Port      Vlans allowed and active in management domain
Fa0/1     10,20,30,40,99
Fa0/2     10,20,30,40,99
Fa0/3     10,20,40,99
Fa0/4     10,20,30,40,99
Gig0/1    10,20,30,40,99

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/1     10,20,30,40,99
Fa0/2     10,20,30,40,99
Fa0/3     10,20,40,99
Fa0/4     10,20,30,40,99
Gig0/1    10,20,30,40,99
```

Figure 17. The key evidence. The injection command (repeated three times, which changes nothing extra), then SW-CORE `show interfaces trunk`: Fa0/3 alone is missing VLAN 30 across all three allowed/active/forwarding lists. Gi0/1 to R1 is trunking normally.

7.7 Step 5 — Apply the smallest fix (and save it)

```
configure terminal
interface fastEthernet0/3
  switchport trunk allowed vlan add 30
end
write memory ! [OK] — the fix is saved; the fault never was
```

`add 30` restores exactly the one VLAN that was removed, without retyping, and so without risking, the rest of the list.

7.8 Step 6 — Verify by repeating the failed tests

CHECK	EXPECTED	OBSERVED
SW-CORE <code>show interfaces trunk</code>	Fa0/3 allows 10,20,30,40,99, same as the others	PASS Fa0/3 back to 10,20,30,40,99
WH-PC-01 → gateway <code>.129</code>	Replies	4/4
WH-PC-01 → FILE-SRV <code>.196</code>	Replies	4/4

```
C:\>ping 10.23.0.129

Pinging 10.23.0.129 with 32 bytes of data:

Reply from 10.23.0.129: bytes=32 time=20ms TTL=255
Reply from 10.23.0.129: bytes=32 time<1ms TTL=255
Reply from 10.23.0.129: bytes=32 time<1ms TTL=255
Reply from 10.23.0.129: bytes=32 time<1ms TTL=255

Ping statistics for 10.23.0.129:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 20ms, Average = 5ms

C:\>ping 10.23.0.196

Pinging 10.23.0.196 with 32 bytes of data:

Reply from 10.23.0.196: bytes=32 time=20ms TTL=127
Reply from 10.23.0.196: bytes=32 time<1ms TTL=127
Reply from 10.23.0.196: bytes=32 time<1ms TTL=127
Reply from 10.23.0.196: bytes=32 time=1ms TTL=127

Ping statistics for 10.23.0.196:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 20ms, Average = 5ms
```

Figure 18. After the fix: WH-PC-01 reaches its gateway (`.129`) and FILE-SRV (`.196`) again, 4/4 each, 0% loss.

Final state. The submitted `tshimo-agri-hq-milestone-2.pkt` was saved after the fix, with `write memory` run on SW-CORE. It contains the healthy, repaired network, not the faulted one. The complete working record is in `troubleshooting/advanced-fault-isolation.md`.

⚠ 8 ISSUES ENCOUNTERED & RESOLUTIONS

Section 8 of 10

These are the real problems hit during implementation, recorded as they happened rather than tidied away.

ISSUE	HOW IT WAS DETECTED	RESOLUTION	WHAT IT SHOWS
Core port plan used Gi0/2–Gi0/5 on SW-CORE	Checking the plan against the 2960-24TT's actual ports before building: it has only two Gigabit ports	Access uplinks moved to Fa0/1–Fa0/4; R1 kept Gi0/1	Port-only change; the VLAN and addressing design was untouched
Like-device links cabled straight-through	Reviewing the build against the rubric's "correct cabling" criterion	ISP–R1 and the four SW-CORE uplinks recabled as crossover; baseline resaved	Correct cabling no longer depends on Auto-MDIX
Native-VLAN-mismatch and spanning-tree "inconsistent VLAN" messages	Console logs on SW-CORE while the switches were configured one at a time	None needed: each cleared once both ends of the trunk had native VLAN 99	A trunk's two ends must agree on its native VLAN
Empty NAT translation table	<code>show ip nat translations</code> run minutes after a successful NAT ping	None needed: PAT entries time out when idle; the successful ping is the proof	Check translations immediately after generating traffic
Wrong config pasted onto the core switch during the fault demo	The Admin control test failed three times in a row (25%, then 100%, then 100% loss), and the core switch's prompt read <code>SW-WH#</code> . <code>show cdp neighbors</code> showed 5 neighbours, so it was physically the core switch. The access-port block from <code>sw-wh.txt</code> had turned three of its trunks (to SW-SAL, SW-WH and SW-SRV) into VLAN 30 access ports, which cut the path to the file server for every department.	Re-applied <code>sw-core.txt</code> in full, confirmed all trunks with <code>show interfaces trunk</code> , saved, then re-injected only the intended one-line fault and re-ran every symptom test	Using CDP to confirm a device's identity, and refusing to accept a symptom that didn't fit the design

Why the last one matters. The first symptom test didn't match the prediction: Admin failed too. Instead of recording it as the result, the mismatch was treated as a second, unknown problem and run down. The symptoms in Section 7 all come from the clean rerun. Recording an unexpected result honestly, then explaining it, is the troubleshooting skill this challenge is about.

9.1 Change Request 1 — implemented on the live network

ITEM	BEFORE CR1	AFTER CR1 (AS BUILT)	CHANGED?
Warehouse staff / PCs	12	20 (WH-PC-01...20)	Yes — endpoints only
SW-WH access ports in VLAN 30	Fa0/2–Fa0/13	Fa0/2–Fa0/21	Yes — port range only
VLAN 30 subnet	10.23.0.128/26	10.23.0.128/26	No
Gateway / R1 subinterface	10.23.0.129, Gi0/0.30	10.23.0.129, Gi0/0.30	No
Trunks, routing, NAT	—	—	No
DHCP WH-POOL	50 leases	50 leases (20 used)	No

CR1 added eight people without any redesign. The /26 already had room for 62 hosts, and the 24-port SW-WH already had the ports. Figure 5 shows the 20 Warehouse ports live in VLAN 30.

9.2 Branch office — still ready

Nothing built in Milestone 2 uses any address in `10.23.128.0/17`, which stays reserved for the branch. All HQ addresses, including the WAN /30 and management /28, sit inside `10.23.0.0/23`, so HQ can still be advertised to a branch as a single summary route. The NAT ACL and the ISP configuration also leave that space untouched.

Repository: **tshimo-agri-network** — <https://github.com/MALULEKE-KS/tshimo-agri-network>

```
tshimo-agri-network/
  README.md          overview, milestone status, document index
  docs/              living design and implementation record
    01-05            Milestone 1 design (requirements → change requests)
    06-project-status-and-milestone-2-control.md
    07-physical-build-design.md    cable schedule, Gate B record
    08-milestone-2-execution-map.md
    09-packet-tracer-build-walkthrough.md
    10-test-results.md    the 12-test log
  configs/           IOS command file per device + server setup
  packet-tracer/      baseline (unconfigured) and milestone-2 (final) .pkt
  screenshots/        20 evidence captures, numbered by test
  troubleshooting/    advanced-fault-isolation.md
  submissions/
    milestone-1/       frozen M1 package
    milestone-2/       frozen M2 package: this report + .pkt
```

10.1 Milestone 2 commit history

COMMIT	WHAT IT ADDED
Add Milestone 2 execution map and Packet Tracer build walkthrough	Phase plan, rubric map, build guide, repository hygiene
Draft IOS configurations for switches, R1 and ISP, plus server setup	All device command files, written and reviewed before being applied
Add baseline Packet Tracer topology and physical build evidence	Unconfigured build, topology screenshots, Gate B record
Milestone 2 implementation, testing and fault-isolation evidence	Configured network, 20 screenshots, test log, troubleshooting record, this report

PROJECT TIMELINE

● Commencement	14 Aug 2026	Complete
● Milestone 1 — Design Review	28 Aug 2026	Design package complete
● Milestone 2 — Implementation Review	2 Oct 2026	Built, tested, fault demonstrated
○ Final Submission	16 Oct 2026	Technical report & video to follow

DECLARATION

I declare that this implementation and its accompanying documentation are my own work, produced for CMPG 325 at North-West University, Mahikeng campus, in accordance with the NWU AI Policy. I built and configured the network in Cisco Packet Tracer, ran every test and captured every screenshot shown in this document, and I understand and can defend every configuration and troubleshooting step described here.

Signature — Kurhula Success Maluleke

2 October 2026

Date

These are the exact command files pasted into each device's CLI, reproduced from `configs/`. Lines starting with `!` are comments that IOS ignores. Server settings are made in Packet Tracer's GUI and are given in Section 5.

SW-CORE — configs/sw-core.txt

```
! =====
! SW-CORE — Tshimo Agri Supplies HQ — Layer 2 distribution switch
! Cisco 2960-24TT
! Role: trunks R1 (Gi0/1) down to the four access switches (Fa0/1-4).
!
!   Holds the full VLAN database since every VLAN passes through here.
! Paste this block into SW-CORE's CLI after Gate B (physical build) passes.
! =====

enable
configure terminal
hostname SW-CORE
no ip domain-lookup

! --- VLAN database ---
vlan 10
  name ADMIN
vlan 20
  name SALES
vlan 30
  name WAREHOUSE
vlan 40
  name SERVERS
vlan 99
  name MGMT-NATIVE
exit

! --- Uplink to R1 (802.1Q trunk, native VLAN 99) ---
interface gigabitEthernet0/1
  description Uplink to R1 Gi0/0
  switchport mode trunk
  switchport trunk native vlan 99
  switchport trunk allowed vlan 10,20,30,40,99
  no shutdown
exit

! --- Trunks to the four access switches ---
interface range fastEthernet0/1 - 4
  switchport mode trunk
  switchport trunk native vlan 99
  switchport trunk allowed vlan 10,20,30,40,99
  no shutdown
exit

interface fastEthernet0/1
  description Trunk to SW-ADM Fa0/1
exit
interface fastEthernet0/2
  description Trunk to SW-SAL Fa0/1
exit
interface fastEthernet0/3
  description Trunk to SW-WH Fa0/1
exit
interface fastEthernet0/4
  description Trunk to SW-SRV Fa0/1
exit

! --- Management SVI on the native VLAN ---
interface vlan 99
  ip address 10.23.1.18 255.255.255.240
  no shutdown
exit

ip default-gateway 10.23.1.17

end
write memory

! -----
! If Packet Tracer's 2960 rejects "switchport mode trunk" with an
! encapsulation error, add this line before it on each trunk
! interface:
!   switchport trunk encapsulation dot1q
! (Not expected on a real 2960-24TT — it's 802.1Q-only — but PT model
```

```
! behaviour can vary; report the exact error if you see one.)
! -----
```

```
! Verify with:
!   show vlan brief
!   show interfaces trunk
!   show ip interface brief
```

SW-ADM — configs/sw-adm.txt

```
! =====
! SW-ADM — Tshimo Agri Supplies HQ — Admin access switch
! Cisco 2960-24TT
! Role: 8 Admin PCs on VLAN 10, trunked back to SW-CORE Fa0/1.
! Paste this block into SW-ADM's CLI after Gate B (physical build) passes.
! =====

enable
configure terminal
hostname SW-ADM
no ip domain-lookup

! --- VLANs used locally (full DB lives on SW-CORE) ---
vlan 10
  name ADMIN
vlan 99
  name MGMT-NATIVE
exit

! --- Uplink to SW-CORE (802.1Q trunk, native VLAN 99) ---
interface fastEthernet0/1
  description Uplink to SW-CORE Fa0/1
  switchport mode trunk
  switchport trunk native vlan 99
  switchport trunk allowed vlan 10,20,30,40,99
  no shutdown
exit

! --- Access ports: 8 Admin PCs ---
interface range fastEthernet0/2 - 9
  switchport mode access
  switchport access vlan 10
  spanning-tree portfast
  no shutdown
exit

! --- Management SVI on the native VLAN ---
interface vlan 99
  ip address 10.23.1.19 255.255.255.240
  no shutdown
exit

ip default-gateway 10.23.1.17

end
write memory

! Verify with:
!   show vlan brief
!   show interfaces trunk
!   show ip interface brief
```

SW-SAL — configs/sw-sal.txt

```
! =====
! SW-SAL — Tshimo Agri Supplies HQ — Sales access switch
! Cisco 2960-24TT
! Role: 12 Sales PCs on VLAN 20, trunked back to SW-CORE Fa0/2.
! Paste this block into SW-SAL's CLI after Gate B (physical build) passes.
! =====

enable
configure terminal
hostname SW-SAL
no ip domain-lookup
```

```
! --- VLANs used locally (full DB lives on SW-CORE) ---
vlan 20
  name SALES
vlan 99
  name MGMT-NATIVE
exit
```

```
! --- Uplink to SW-CORE (802.1Q trunk, native VLAN 99) ---
interface fastEthernet0/1
  description Uplink to SW-CORE Fa0/2
  switchport mode trunk
  switchport trunk native vlan 99
  switchport trunk allowed vlan 10,20,30,40,99
  no shutdown
exit
```

```
! --- Access ports: 12 Sales PCs ---
interface range fastEthernet0/2 - 13
  switchport mode access
  switchport access vlan 20
  spanning-tree portfast
  no shutdown
exit
```

```
! --- Management SVI on the native VLAN ---
interface vlan 99
  ip address 10.23.1.20 255.255.255.240
  no shutdown
exit
```

```
ip default-gateway 10.23.1.17
```

```
end
write memory
```

```
! Verify with:
!   show vlan brief
!   show interfaces trunk
!   show ip interface brief
```

SW-WH — configs/sw-wh.txt

```
! =====
! SW-WH – Tshimo Agri Supplies HQ – Warehouse access switch
! Cisco 2960-24TT
! Role: 20 Warehouse PCs (12 existing + 8 CR1) on VLAN 30,
!       trunked back to SW-CORE Fa0/3.
! Paste this block into SW-WH's CLI after Gate B (physical build)
! passes.
!
! NOTE: SW-CORE's Fa0/3 is the trunk port used for the assigned
! troubleshooting fault (VLAN 30 removed from its allowed list).
! This file configures the healthy baseline only – the fault is
! injected later, on SW-CORE, per troubleshooting/advanced-fault-
! isolation.md.
! =====
```

```
enable
configure terminal
hostname SW-WH
no ip domain-lookup
```

```
! --- VLANs used locally (full DB lives on SW-CORE) ---
vlan 30
  name WAREHOUSE
vlan 99
  name MGMT-NATIVE
exit
```

```
! --- Uplink to SW-CORE (802.1Q trunk, native VLAN 99) ---
interface fastEthernet0/1
  description Uplink to SW-CORE Fa0/3
  switchport mode trunk
  switchport trunk native vlan 99
  switchport trunk allowed vlan 10,20,30,40,99
  no shutdown
exit
```

```
! --- Access ports: 20 Warehouse PCs (includes CR1 growth) ---
interface range fastEthernet0/2 - 21
  switchport mode access
  switchport access vlan 30
  spanning-tree portfast
  no shutdown
exit
```

```
! --- Management SVI on the native VLAN ---
interface vlan 99
  ip address 10.23.1.21 255.255.255.240
```

```
no shutdown
exit

ip default-gateway 10.23.1.17

end
write memory

! Verify with:
!   show vlan brief
!   show interfaces trunk
!   show ip interface brief
```

SW-SRV — configs/sw-srv.txt

```
! =====
! SW-SRV – Tshimo Agri Supplies HQ – Server access switch
! Cisco 2960-24TT
! Role: DHCP-SRV, DNS-SRV, FILE-SRV on VLAN 40, trunked back to SW-
! CORE Fa0/4.
! Paste this block into SW-SRV's CLI after Gate B (physical build)
! passes.
! =====
```

```
enable
configure terminal
hostname SW-SRV
no ip domain-lookup
```

```
! --- VLANs used locally (full DB lives on SW-CORE) ---
vlan 40
  name SERVERS
vlan 99
  name MGMT-NATIVE
exit
```

```
! --- Uplink to SW-CORE (802.1Q trunk, native VLAN 99) ---
interface fastEthernet0/1
  description Uplink to SW-CORE Fa0/4
  switchport mode trunk
  switchport trunk native vlan 99
  switchport trunk allowed vlan 10,20,30,40,99
  no shutdown
exit
```

```
! --- Access ports: DHCP-SRV, DNS-SRV, FILE-SRV ---
interface range fastEthernet0/2 - 4
  switchport mode access
  switchport access vlan 40
  spanning-tree portfast
  no shutdown
exit
```

```
! --- Management SVI on the native VLAN ---
interface vlan 99
  ip address 10.23.1.22 255.255.255.240
  no shutdown
exit
```

```
ip default-gateway 10.23.1.17
```

```
end
write memory
```

```
! Verify with:
!   show vlan brief
!   show interfaces trunk
!   show ip interface brief
```

R1 — configs/r1.txt

```
! =====
! R1 – Tshimo Agri Supplies HQ – edge router
! Cisco 2911
! Role: router-on-a-stick inter-VLAN routing on Gi0/0 (trunk to SW-
! CORE Gi0/1),
!       DHCP relay to DHCP-SRV, WAN to ISP on Gi0/1, NAT/PAT
!       overload.
! Paste AFTER the five switch configs are in.
! =====
```

```
enable
configure terminal
hostname R1
no ip domain-lookup
```

```
! --- Physical trunk interface: no IP, just bring it up ---
interface gigabitEthernet0/0
```

```

description Trunk to SW-CORE Gi0/1
no ip address
no shutdown
exit

! --- One subinterface per VLAN = that VLAN's default gateway ---
interface gigabitEthernet0/0.10
description VLAN 10 ADMIN gateway
encapsulation dot1Q 10
ip address 10.23.0.1 255.255.255.192
ip helper-address 10.23.0.194
ip nat inside
exit

interface gigabitEthernet0/0.20
description VLAN 20 SALES gateway
encapsulation dot1Q 20
ip address 10.23.0.65 255.255.255.192
ip helper-address 10.23.0.194
ip nat inside
exit

interface gigabitEthernet0/0.30
description VLAN 30 WAREHOUSE gateway
encapsulation dot1Q 30
ip address 10.23.0.129 255.255.255.192
ip helper-address 10.23.0.194
ip nat inside
exit

interface gigabitEthernet0/0.40
description VLAN 40 SERVERS gateway
encapsulation dot1Q 40
ip address 10.23.0.193 255.255.255.192
ip nat inside
exit

! Native VLAN 99: untagged frames on the trunk belong here
interface gigabitEthernet0/0.99
description VLAN 99 MGMT-NATIVE gateway
encapsulation dot1Q 99 native
ip address 10.23.1.17 255.255.255.240
exit

! --- WAN to ISP ---
interface gigabitEthernet0/1
description WAN to ISP
ip address 10.23.1.1 255.255.255.252
ip nat outside
no shutdown
exit

! --- Default route out to the ISP ---
ip route 0.0.0.0 0.0.0.0 10.23.1.2

! --- NAT/PAT: translate the business/server range (10.23.0.0/24)

```

```

only.
!      WAN and management (10.23.1.0/24) are deliberately excluded. -
--
access-list 1 permit 10.23.0.0 0.0.0.255
ip nat inside source list 1 interface gigabitEthernet0/1 overload

end
write memory

! Verify with:
!   show ip interface brief      (all subinterfaces + Gi0/1 up/up)
!   show ip route                (C routes for each VLAN, S*
!   default to 10.23.1.2)
!   show ip nat translations     (after a PC pings 203.0.113.1)

```

ISP — configs/isp.txt

```

! =====
! ISP – simulated upstream provider
! Cisco 2911
! Role: far end of the R1 WAN /30, plus a loopback acting as an
!       "internet" test target for NAT verification.
!
! ISP port facing R1 Gi0/1 is Gi0/0. Confirmed 2 Oct 2026: this file
! was
! applied unchanged, R1 Gi0/1 came up/up, and a PC reached
! 203.0.113.1
! through NAT – impossible unless the link lands on the addressed
! Gi0/0.
! =====

enable
configure terminal
hostname ISP
no ip domain-lookup

interface gigabitEthernet0/0
description WAN to Tshimo R1 Gi0/1
ip address 10.23.1.2 255.255.255.252
no shutdown
exit

! Simulated internet host
interface loopback0
description Internet test target
ip address 203.0.113.1 255.255.255.255
exit

end
write memory

! No route back to 10.23.0.0/24 is needed: R1 NATs every inside
! host to 10.23.1.1, which the ISP already reaches directly.

```